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# Time-resolved spectroscopy uncovers deprotonation-induced reconstruction in oxygen-evolution NiFe-based (oxy) hydroxides

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Transition-metal layered double hydroxides are widely utilized as electrocatalysts for the oxygen evolution reaction (OER), undergoing dynamic transformation into active oxyhydroxides during electrochemical operation. Nonetheless, our understanding of the non-equilibrium structural changes that occur during this process remains limited. In this study, utilizing in situ energy-dispersive X-ray absorption spectroscopy and machine learning analysis, we reveal the occurrence of deprotonation and elucidate the role of incorporated iron in facilitating the transition from nickel-iron layered double hydroxide (NiFe LDH) into its active oxyhydroxide. Our findings demonstrate that iron substitution promotes deprotonation process within NiFe LDH, resulting in the preferential removal of protons from the specific bridged hydroxyl group ($Ni^{2+}$-OH-$Fe^{3+}$) linked to edge-sharing [$NiO_6$] and [$FeO_6$] octahedron. This deprotonation behavior drives the formation of high-valence $Ni^{3+\delta}$ species ($0 < \delta < 1$), which subsequently serve as the active sites, thereby ensuring efficient oxygen evolution activity. This approach offers high-resolution insights of dynamic structural evolution, overcoming the limitations of extended acquisition times and advancing our understanding of OER mechanisms.

Drawing a comprehensive picture towards the evolution of active sites during oxygen evolution reaction (OER) is essential for elucidating origins of catalytic activity and advancing the development of efficient catalysts[1–3]. Layered double hydroxides (LDHs) composed of first-row transition metals have emerged as prominent OER catalysts due to their favorable performance, tunable chemical compositions, and abundance of active sites[4–6]. Particularly, NiFe LDHs, which consist of [$NiO_6$] and substituted [$FeO_6$] octahedrons, exhibit superior OER activity, garnering intense attention in recent years[7–11]. Remarkably, the geometric and electronic structures of NiFe LDHs have been observed to undergo self-reconstruction under applied potentials, resulting in improved OER performance compared to unmodified $Ni(OH)_2$[3,9,12,13]. Nevertheless, the specific impact of incorporated Fe heteroatoms remains unclear, primarily due to insufficient direct understanding of the dynamic

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interactions between Ni and Fe species under OER conditions[14–17]. Consequently, real-time monitoring of OER-related behavior on NiFe LDHs, with special focus on the evolution of local coordination environment, is necessary to gain deeper insights into catalytic process[18].

In this regard, in situ characterization techniques, particularly structure-sensitive X-ray absorption fine structure (XAFS) spectroscopy, have been widely employed to explore the evolution of active sites in NiFe LDHs. Additionally, within [Ni/$FeO_6$] octahedrons, elevated valence states of Ni sites and shortened Ni/Fe-O bonds were observed, providing evidence for the formation of active Ni(Fe)OOH during OER[8,19–21]. More importantly, it is widely acknowledged that the evolution of active species is closely associated with the deprotonation behavior, which involves the removal of protons surrounding the [Ni/$FeO_6$] octahedrons[22–25]. Despite this consensus, our understanding of the transient deprotonation remains limited, particularly in identifying key post-deprotonation intermediates (Fig. 1a). The rapid nature of deprotonation, coupled with the constraints of commonly used techniques, which are primarily proficient in capturing limited structural information in equilibrium states, hinder our ability to provide robust evidence for the occurrence of this process. Therefore, there is a pressing need to explore advanced time-resolved techniques that offer deeper insights into the deprotonation, thus uncovering the origin of OER activity in NiFe LDHs[18]. Compared to standard step-scanning XAFS, energy-dispersive XAFS (ED-XAFS), equipped with a curved polychromator and position-sensitive detector (PSD), is expected to address this challenge because it has the capability to acquire the entire spectrum in a single-shot recording[26–28].

In this work, we utilize in situ electrocatalysis-fit ED-XAFS, machine learning, and theoretical calculations to conduct a time-resolved investigation into the deprotonation of NiFe LDH under OER conditions, revealing the promoting role of Fe in this process. Our findings reveal that the Fe substitution enables the preferential removal of protons from the bridged $OH_{Ni\text{-}O\text{-}Fe}$ groups positioned between [$NiO_6$] and [$FeO_6$] octahedrons. This preferential deprotonation behavior leads to significant structural distortion of the [Ni/$FeO_6$] octahedrons and concomitantly enhances the accumulation of high-valence $Ni^{3+\delta}$ ($0 < \delta < 1$) species. Importantly, the dynamically generated $Ni^{3+\delta}$ species subsequently serve as active sites, weakening the binding strength of intermediates and enhancing OER activity, as rationalized by the theoretical calculations.

## Results and discussion

Catalysts of $Ni(OH)_2$ with varying degrees of Fe substitution were successfully prepared, as confirmed by X-ray diffraction (XRD) patterns (Supplementary Fig. 1). Subsequently, linear sweep voltammetry (LSV) measurements reveal that the $Ni_{0.83}Fe_{0.17}$ LDH, doped with 17 wt% Fe, exhibits notable OER activity, achieving an overpotential of 235 mV at a current density of 10 mA cm$^{-2}$ (Fig. 1b). Furthermore, the $Ni_{0.95}Fe_{0.05}$ and $Ni_{0.83}Fe_{0.17}$ catalysts demonstrate more pronounced oxidation peaks of Ni species and higher OER activity than those of $Ni(OH)_2$,

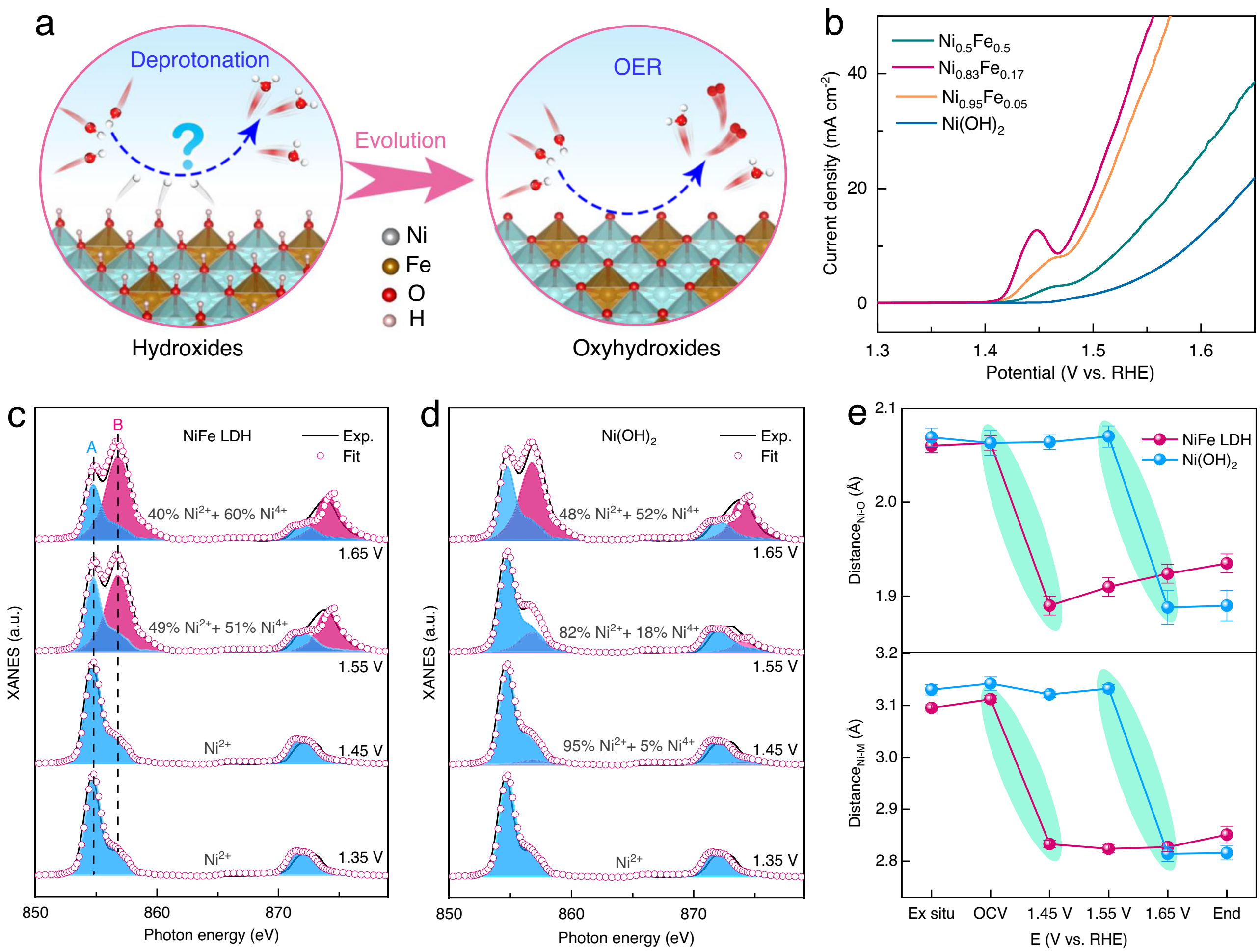


**Fig. 1 | Electrochemical measurement and characterization.** **a** Schematic of the evolution of hydroxides into oxyhydroxides during OER. **b** LSV curves of different Fe doped $Ni(OH)_2$ catalysts. Ni L-edge sXAS of (**c**) NiFe LDH and (**d**) $Ni(OH)_2$. The characteristic L-edge peaks of NiO and $K_2Ni(H_2IO_6)_2$ were used as references for $Ni^{2+}$ and $Ni^{4+}$, respectively, to illustrate the mixed oxidation states present in the $Ni^{3+\delta}$ of NiFe LDH and $Ni(OH)_2$ under OER conditions. The term of "a.u." denotes arbitrary units. **e** Changes of bond length of Ni-O and Ni-M coordination shells for NiFe LDH and $Ni(OH)_2$ at different OER conditions.

highlighting the substantial impact of Fe substitution on OER behavior. However, further increasing Fe doping to 50 wt% results in a decline in OER activity due to the phase separation by excess Fe. To elucidate the fundamental role of Fe, a comparative analysis was conducted between $Ni_{0.83}Fe_{0.17}$ (hereafter NiFe LDH) and $Ni(OH)_2$. Both catalysts exhibit comparable ultrathin nanosheet morphologies (Supplementary Fig. 2), ensuring the exposure of abundant active sites. X-ray absorption near edge structure (XANES) spectroscopy reveals that the average valence states of Ni and Fe closely resemble $Ni^{2+}$ and $Fe^{3+}$, respectively, as evidenced by the positions of their absorption edges ($E_0$) (Supplementary Figs. 3 and 4). Furthermore, the fitting of spectra indicates a comparable local coordination environment between NiFe LDH and $Ni(OH)_2$ (Supplementary Fig. 5 and Table 1)[29].

In situ XAFS measurements were performed to compare the potential-dependent oxidation behaviors between NiFe LDH and $Ni(OH)_2$. Both catalysts deliver a high-energy shift in $E_0$ with increasing anodic potentials, indicating an elevated oxidation state in Ni species (Supplementary Figs. 6 and 7). Interestingly, unlike $Ni(OH)_2$, NiFe LDH shows accelerated oxidation of $Ni^{2+}$ species, evident from a significant 3 eV shift in $E_O$ observed between 1.45 and 1.55 V *vs.* RHE (Supplementary Fig. 8). In comparison, $Ni(OH)_2$ does not exhibit noticeable $Ni^{3+\delta}$ formation until the potential reaches 1.65 V. Additionally, valence-sensitive soft X-ray absorption spectroscopy (sXAS) confirms this accelerated oxidation (Fig. 1c, d, and Supplementary Figs. 9–11)[13,30]. Notably, at 1.55 V, the percentage of high-valence $Ni^{3+\delta}$ species in NiFe LDH increases to 51%, whereas a higher potential of 1.65 V is required for $Ni(OH)_2$.

This hysteresis is also reflected in the local atomic structure. Compared to open-circuit voltage (OCV) states, two coordination peaks in Ni and Fe K-edge FT-EXAFS spectra exhibit noticeable low-R shifts, suggesting a decrease in bond distances for both catalysts (Supplementary Figs. 12–15). Combining quantitative fitting parameters with postmortem characterizations provides additional evidence of an irreversible potential-driven transformation from hydroxide to oxyhydroxide, accompanied by a contraction of octahedrons (Supplementary Figs. 16–19 and Tables 2–4). Intriguingly, a notable hysteresis in the required potentials is evident for $Ni(OH)_2$, where the contraction of $[NiO_6]$ octahedron occurs at 1.65 V, while NiFe LDH experiences this transformation at a lower voltage of 1.45 V (Fig. 1e). These observations indicate that the incorporation of Fe species effectively promotes the oxidation of $Ni^{2+}$ into $Ni^{3+\delta}$.

The XAFS results presented in Fig. 1e reveal the phase transformation from NiFe LDH to Ni(Fe)OOH. However, even with narrower voltage intervals applied during in situ probing, capturing the richer intermediates involved in the structural transition of NiFe LDH remains challenging. Therefore, to gain insight into the non-equilibrium changes during this transition, we resorted to an ED-XAFS technique with a 60-millisecond resolution, allowing us to effectively track the time-dependent structural evolution at a specific voltage. In the ED-XAFS setup, a polychromatic X-ray beam is dispersed and focused onto an electrochemical cell by a curved polychromator (Fig. 2a). During the measurement, the electrolyte was kept flowing to dissipate the heat load caused by the dispersed X-rays (Supplementary Figs. 20 and 21). The PSD then records the entire spectrum by acquiring all data points simultaneously in a single shot, enabling the detection of time-resolved changes[27,31]. At first glance, a slight variation is detected in the first 1000 ED-XAFS spectra acquired within 60 s, as indicated by the decreased intensity of white-line peak (Supplementary Figs. 22 and 23). To further amplify and observe this change, we continuously recorded 10,000 XANES spectra, as depicted in Fig. 2b. Clearly, $E_O$ exhibits a high-energy shift and is accompanied by a decreased intensity of white-line peak, suggesting an increase in the Ni valence state and a modification in local structure (Fig. 2b and Supplementary Fig. 24). These changes become even more apparent when examining the difference curves of XANES spectra, highlighting variations within NiFe LDH (Fig. 2c).

To further distinguish the possible intermediate components involved in this evolution, a machine learning analysis incorporating principal component analysis (PCA) and linear combination analysis (LCA) methods was performed (Supplementary Note 1)[32,33]. During this structural evolution, two principal components (PCs) were identified through PCA approach (Fig. 2d). Furthermore, LCA was performed to track the evolution trend of two intermediate components, defined as $C_1$ and $C_2$. The XANES spectra of NiFe LDH and Ni(Fe)OOH were used as references in the analysis of a total of 10,000 unknown spectra (Supplementary Fig. 25). LCA results in Fig. 2e demonstrate a gradual decrease in the relative concentration of $C_1$, while the $C_2$ shows a significant increase. This attribution is further supported by the comparison of the LCA-fitting XANES spectra with the experimental spectra (Supplementary Fig. 26). The reconstruction from hydroxide and oxyhydroxide is also evident in the data-driven analysis of the Fe K-edge of NiFe LDH (Supplementary Fig. 27). The ED-XAFS spectra of unmodified $Ni(OH)_2$ exhibit a decrease in the white-line peak at the Ni K-edge, accompanied by a high-energy shift of $E_O$ during the reaction, indicating the oxidation of $Ni^{2+}$ species, consistent with observations in NiFe LDH. Additionally, PCA of the ED-XAFS results for $Ni(OH)_2$ reveals two principal components, with a relatively small eigenvalue contribution from PC2 (Supplementary Fig. 28). While LCA suggests a two-phase transition in the reconstruction of $Ni(OH)_2$, the variation trend diverges from that of NiFe LDH. During the transformation of $Ni(OH)_2$, a temporary plateau is observed, possibly due to limited deprotonation hindered by lattice disruption that affects H atom diffusion[22]. In contrast, NiFe LDH exhibits a continuous linear transformation, suggesting that Fe enhances structural reconstruction, promoting the formation of high-valence $Ni^{3+\delta}$ species. This finding aligns with the XAFS and sXAS results.

Previous studies have suggested a potential connection between this transition and the potential-driven deprotonation process of $[Ni/FeO_6]$ octahedrons, but experimental evidence remains lacking[22–25,34]. Since the H atom resides in the second coordination shell of the metal site and exhibits weak scattering of photoelectrons, directly observing a dynamic removal of H from the $[Ni/FeO_6]$ octahedra is difficult using both ED-XAFS and standard XAFS. Consequently, we employ an indirect approach to explore the role of incorporated Fe in deprotonation behavior (Supplementary Fig. 29). We first resorted to density functional theory (DFT) calculations to determine the underlying deprotonation-induced intermediates. Specifically, in NiFe LDH, the substitution of Fe for Ni sites results in two distinct types of OH groups: the pristine $OH_{Ni\text{-}O\text{-}Ni}$ and the newly formed bridged $OH_{Ni\text{-}O\text{-}Fe}$, which is coordinated to both Ni and Fe centers (Fig. 3a). To assess the influence of substituting Fe on deprotonation behaviors in NiFe LDH, a 2 × 2 supercell was constructed, containing four metal centers with two $OH_{Ni\text{-}O\text{-}Ni}$ and six $OH_{Ni\text{-}O\text{-}Fe}$ groups. This model highlights the comparable local coordination environments of NiFe LDH while simplifying the analysis by disregarding the contribution of intercalated species.

Interestingly, a favorable deprotonation is observed in $OH_{Ni\text{-}O\text{-}Fe}$, exhibiting a notably lower deprotonation energy of 1.13 eV, in contrast to pristine $OH_{Ni\text{-}O\text{-}Ni}$ (2.0 eV), consistent with reported findings[23]. Furthermore, we investigated the entire deprotonation process from NiFe LDH to Ni(Fe)OOH by considering the dynamic adsorption and desorption of protons. The stepwise energies and corresponding lattice constants are presented in Fig. 3b. Specifically, the removal of the four protons preferentially starts from the bridged $OH_{Ni\text{-}O\text{-}Fe}$, followed by the detachment of one proton from pristine $OH_{Ni\text{-}O\text{-}Ni}$. Subsequently, the remaining protons are sequentially removed from both the bridged $OH_{Ni\text{-}O\text{-}Fe}$ and pristine $OH_{Ni\text{-}O\text{-}Ni}$, concluding the deprotonation process with the removal of the last H linked to bridged $OH_{Ni\text{-}O\text{-}Fe}$ (Fig. 3c). Importantly, the total deprotonation energy for NiFe LDH is calculated to be 14.93 eV, whereas a higher energy of 20.27 eV is

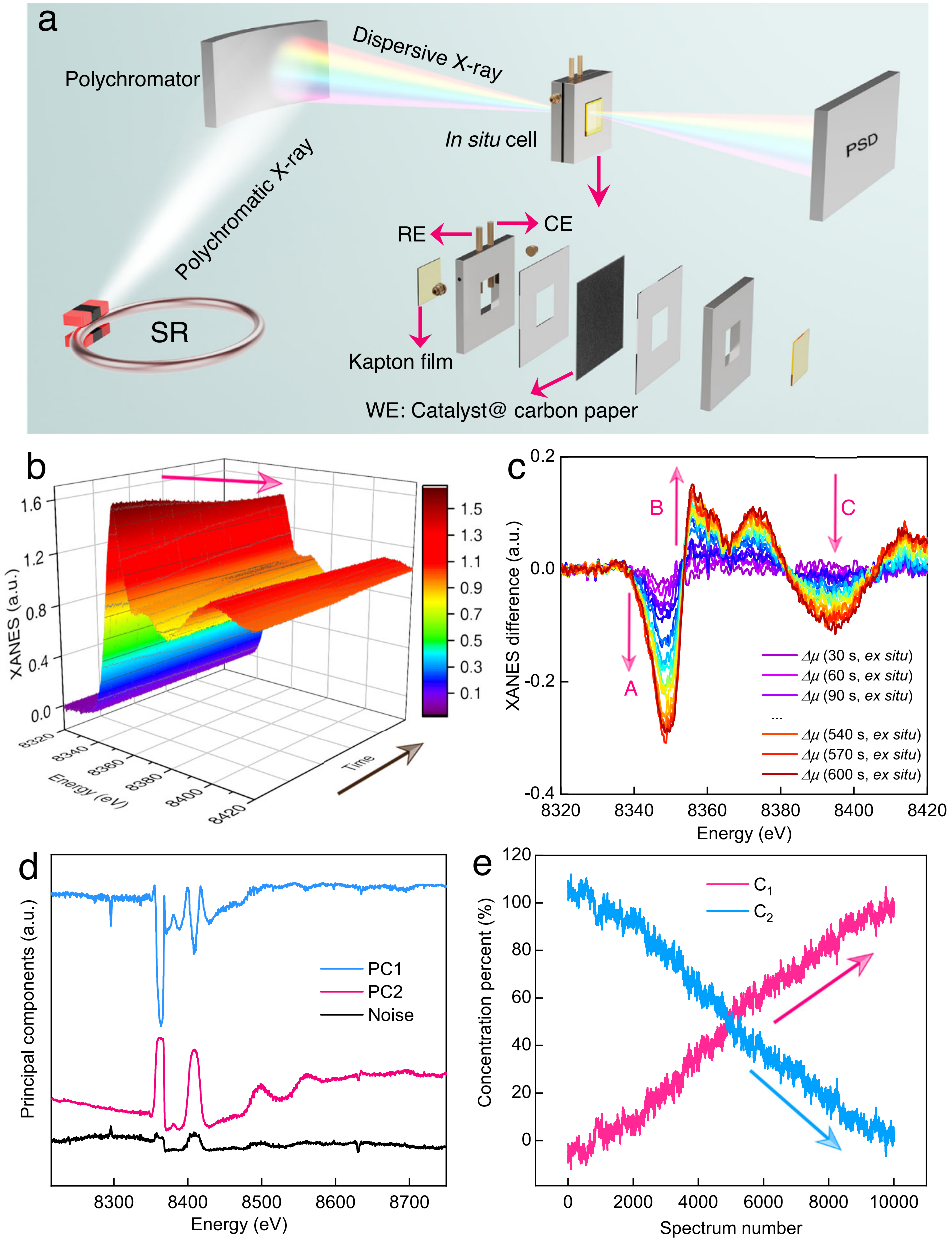


**Fig. 2 | Time-resolved ED-XAFS measurements coupled with machine learning analysis. a** Schematic diagram of in situ ED-XAFS setup, in which "SR", "WE", "CE" and "RE" represent "synchrotron radiation facility", "working electrode", "counter electrode" and "reference electrode", respectively. **b** Ni K-edge XANES spectra of NiFe LDH recorded using ED-XAFS. **c** Extracted XANES difference spectra. **d** PCA results extracted from ED-XAFS spectra. **e** The relative concentrations evolution obtained from LCA.

required for $Ni(OH)_2$ (Supplementary Fig. 30). This indicates that the Fe substitution facilitates the deprotonation behavior. Additionally, the equilibrium phase diagram demonstrates that the theoretical potential for this deprotonation-induced transition is 1.56 V (Supplementary Fig. 31), which can be achieved with the applied potential.

Based on these deprotonated-induce structures, XANES simulations were further performed to produce simulated XANES spectra for the post-deprotonation intermediates (Supplementary Fig. 29). By comparing characteristic peaks, particularly the white-line (A) and multiple scattering peaks (B and C), it is evident that the calculated spectra can accurately reproduce the main features observed in experimental ED-XAFS spectra (Fig. 3d). Notably, these simulated characteristic peaks deliver a significant high-energy shift towards Ni(Fe)OOH. Additionally, the structural alterations resulting from the H atoms removal are clearly identifiable, affirming that this structural transformation occurs via the deprotonation. In this regard, millisecond-resolved ED-XAFS can obtain richer information about post-deprotonation intermediates than standard XAFS. This enhanced capability of ED-XAFS allows more accurate confirmation of the occurrence of deprotonation by integrating XANES simulations with DFT calculations. Consequently, ED-XAFS analysis highlights the crucial role of Fe in facilitating the deprotonation process. Specifically, unlike the initial $Ni^{2+}$-OH-$Ni^{2+}$, Fe substitution results in the formation of a specific bridged $Ni^{2+}$-OH-$Fe^{3+}$ group. This configuration enables selective and preferential proton removal, thereby promoting the transformation of NiFe LDH into the OER-active Ni(Fe)OOH.

To understand the mechanism behind enhanced OER activity, we further investigate the electronic information altered by deprotonation behavior. Initially, near the Fermi level ($E_F$) of NiFe LDH, there is an obvious band gap of 2.37 eV, coupled with a lack of electronic interaction, which indicates poor electrical conductivity (Fig. 4a)[25]. In contrast, upon proton extraction, Ni(Fe)OOH exhibits a narrowed band

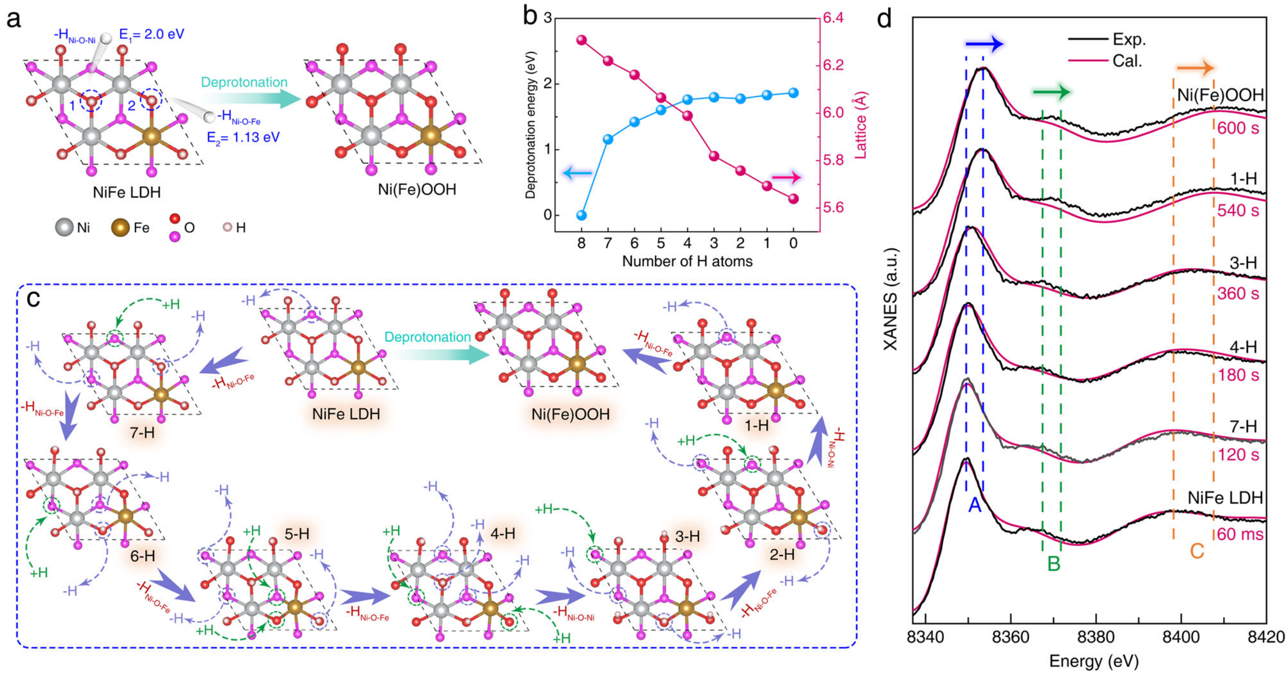


**Fig. 3 | Theoretical calculations and XANES simulations investigate the deprotonation process. a** Illustration of the different OH groups formed in NiFe LDH and corresponding deprotonation energies after Fe substitution. The orientations of H atoms are distinguished with different colored balls, where red and pink balls represent the out-of-plane and in-plane H atoms, respectively. **b** Energies required to remove all H atoms from NiFe LDH and corresponding lattice constant. **c** Entire deprotonation process of NiFe LDH revealed by theoretical calculations. **d** Comparison of experimental and calculated ED-XAFS spectra for Ni K-edge. From bottom to top, the presented ED-XAFS spectra correspond to sequence numbers 1, 2000, 3000, 6000, 9000, and 10,000, respectively.

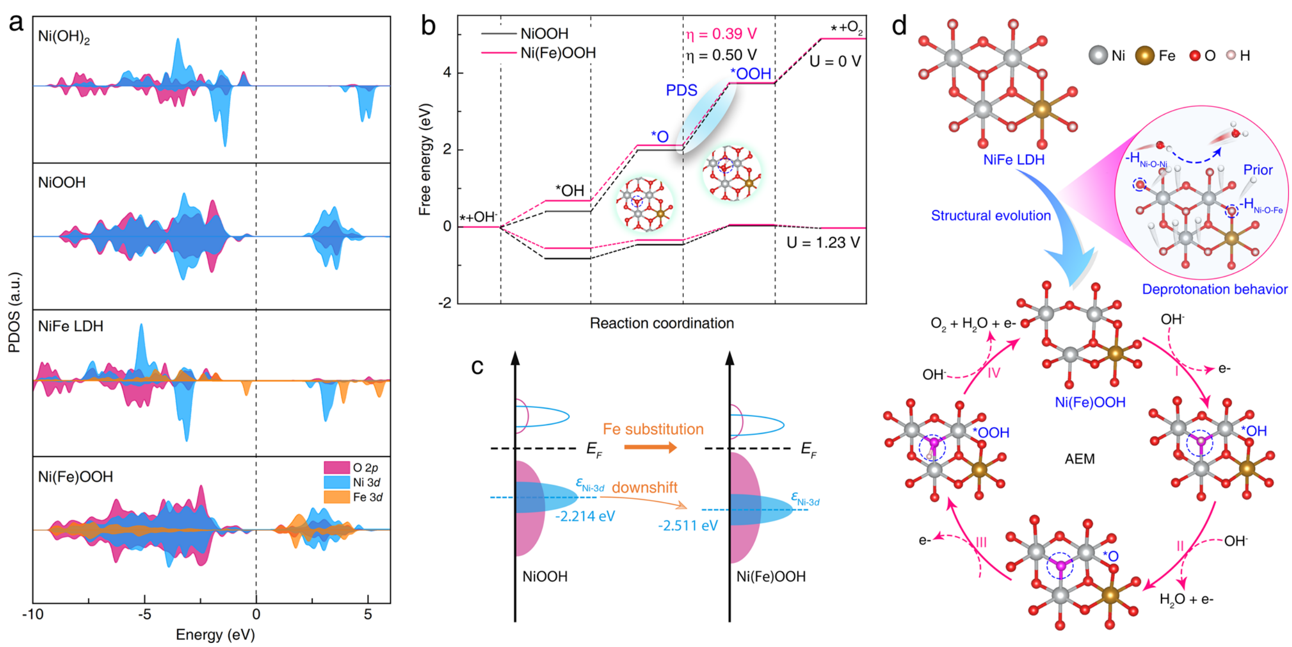


**Fig. 4 | Investigation of the enhanced OER activity. a** Projected density of states (PDOSs) plots of initial $Ni(OH)_2$, NiFe LDH, and deprotonated NiOOH, Ni(Fe)OOH. **b** Free energy diagram for alkaline OER. **c** Schematic diagram of the changes in electronic structure resulting from Fe substitution. **d** Schematic of the enhanced mechanism of NiFe LDH on alkaline OER.

gap with a strong overlap between the O 2*p* and metal 3 *d* orbitals. This suggests that the dynamically generated active site, resulting from the deprotonation process, favors electrocatalytic reaction. Similarly, a comparable improvement is observed in $Ni(OH)_2$ upon deprotonation, which further confirms that the deprotonated oxyhydroxides, rather than the initial hydroxides, serve as the active species.

Based on the combination of EXAFS fitting and O 1*s* X-ray photoelectron spectroscopy results (Supplementary Fig. 32, Tables 2 and 4), we introduced an unsaturated Ni site in the reconstructed oxyhydroxides by removing an oxygen atom. This allowed us to evaluate the OER free energy diagram with the adsorbate evolution mechanism (AEM), as shown in Supplementary Figs. 33–37 and Supplementary Data 1[34–36]. Both

NiOOH and Ni(Fe)OOH exhibit a similar trend in free energy changes (Fig. 4b), with the formation of *OOH from *O identified as the potential-determining step (PDS). NiOOH exhibits a theoretical overpotential of 0.50 V, owing to the excessively strong binding of *O. In contrast, Ni(Fe)OOH requires only 0.39 V of theoretical overpotential to overcome the PDS, implying that the introduction of Fe improves OER activity by weaking the binding affinity of *O. This improvement can be further elucidated by examining changes in *d*-band center energies (Supplementary Table 5). Compared to NiOOH (−2.214 eV), Fe substitution notably shifts the Ni *d*-band center energy to −2.511 eV (Fig. 4c). This downshift ensures weaker metal-oxygen binding, particularly for *O, thus enhancing OER activity. Therefore, the alkaline OER enhanced mechanism can be described for NiFe LDH (Fig. 4d). The initial activation step involves the dynamic evolution from NiFe LDH to OER-active Ni(Fe)OOH, facilitated by a more facile deprotonation process within the $OH_{Ni\text{-}O\text{-}Fe}$ groups. This is followed by a four-electron transfer that releases $O_2$ at the high-valence $Ni^{3+\delta}$ sites.

In summary, our integration of time-resolved ED-XAFS with machine learning analysis provides additional experimental insights into the optimized mechanism for the transition NiFe LDH to Ni(Fe)OOH. Specifically, the introduction of Fe facilitates the preferential removal of protons from the bridged $OH_{Ni\text{-}O\text{-}Fe}$ group that connects the Ni and Fe centers. Compared to $Ni(OH)_2$, this facilitated deprotonation process promotes the formation of active $Ni^{3+\delta}$ species, thereby effectively boosting OER activity. This work offers a real-time perspective on the origins of enhanced OER activity from self-reconstruction in transition-metal (oxy)hydroxides and guides the investigation of other electrocatalytic processes using a data-driven, time-resolved ED-XAFS approach.

## Methods

### Synthesis of NiFe LDH and $Ni(OH)_2$

To synthesize the NiFe LDH catalyst, nickel acetate tetrahydrate and ferric chloride hexahydrate were mixed in varying molar ratios (total metal molar content: 0.5 mmol) and dissolved in 200 mL of ethanol as the precursor solution. The mixture was stirred at 90 °C to ensure homogeneity. Under vigorous stirring, 2 mL of ammonium hydroxide solution was gradually added dropwise to the solution. Subsequently, 100 mL of deionized (DI) water was introduced into the mixture, which was then continuously stirred at 90 °C for 90 min. The product was collected via centrifugation and washed three times with DI water and ethanol. $Ni(OH)_2$ was synthesized by dissolving 1.5 mmol nickel chloride and 4 mmol urea in 20 mL DI water. The resulting solution was transferred into a Teflon-lined autoclave and heated at 100 °C for 3 h. The product was recovered, cleaned with DI water and ethanol, and subsequently treated with ultrasound in ethanol for 24 h.

### Electrochemical measurements

Electrochemical performance of NiFe LDH and reference catalysts was assessed at room temperature using a CHI760E electrochemistry workstation with a three-electrode configuration. Hg/HgO electrode and carbon rod serve as reference electrode and counter electrode, respectively. To prepare the working electrode, 5.0 mg of catalyst was suspended in a solution comprising 500 µL of ethanol, 500 µL of deionized water, and 40 µL of 5 wt% Nafion solution (Sigma-Aldrich). The suspension was ultrasonicated for 6 h to obtain a uniform ink. Subsequently, 5 µL of the ink was deposited onto a glassy carbon electrode (3 mm diameter) and allowed to dry naturally. 1 M KOH was functioned as electrolyte for electrochemical measurements. The potentials versus RHE were converted using the Eq. (1).

$$E_{(RHE)} = E_{(Hg/HgO)} + 0.059 \times pH + 0.098V \tag{1}$$

### XAFS measurements

XAFS spectra of Ni (8333 eV) and Fe K-edge (7112 eV) were recorded at the BL14W1 beamline of the Shanghai Synchrotron Radiation Facility (SSRF), China. In situ XAFS measurements were conducted in transmission mode using a custom-designed cell. Specifically, the working electrode, consisting of catalyst-loaded carbon paper. Specially, it was prepared by dispersing the catalyst in a solution of 500 µL ethanol, 500 µL deionized water, and 40 µL of 5 wt% Nafion solution. The mixture was ultrasonicated for 6 h to form a uniform ink, which was then applied onto carbon paper secured with polyimide tape.

### Soft XAS measurements

sXAS measurements at the Ni and Fe L-edge were conducted at the BL10B and BL12B beamlines of the National Synchrotron Radiation Laboratory (NSRL), China. Catalyst ink was coated onto carbon paper to fabricate working electrodes, which were maintained at specific potentials for 10 minutes. The post-testing electrodes were then dried and characterized. The sXAS measurements were carried out by total electron yield mode.

### In situ ED-XAFS measurements

Ni and Fe K-edge ED-XAFS spectra were collected at the BL05U beamline of SSRF, China, with a photon flux focused on the sample of approximately $2.5 \times 10^{12}$ phs/s at 7.2 keV. The exposure time was set to 4.2 milliseconds. Due to the limitations of the detector's readout speed, the effective time resolution of ED-XAFS is currently limited to 60 milliseconds. Specifically, the uniformly distributed catalyst ink was drop-casted slowly onto carbon paper by a spray coating machine to ensure sample uniformity. The carbon paper loaded with catalyst was used as working electrode and embedded in a customized cell for ED-XAFS measurement. The position of in situ cell was adjusted to ensure that the dispersive X-ray could accurately focus on the catalyst and could be detected by the PSD in transmission mode. The measurement was carried out under the chronoamperometry mode of 1.65 V *vs*. RHE in 1 M KOH. The collection of spectra and the application of potential start at the same time.

### XAFS data analysis

ATHENA module implemented in the IFEFFIT was used to analyze the recorded XAFS data For Ni and Fe K-edge spectra, $k^3$-weighted $\chi(k)$ data (2.35–12.35 Å$^{-1}$) were Fourier-transformed to R-space. To separate contributions from different coordination shells, a Hanning window ($dk = 1.0$ Å$^{-1}$) was selected. ARTEMIS module with IFEFFIT was used to conduct quantitative curve fitting in R-space for obtaining structural parameters, including Debye-Waller factors ($\sigma^2$), coordination numbers ($CN$), interatomic distances ($R$), and energy shift ($\Delta E_0$). Effective backscattering amplitudes and phase shifts for fitting paths were calculated using the FEFF8.0 code. During fitting, above structural parameters were treated as adjustable parameters for all samples and paths.

### XANES simulations

XANES simulations were performed using the finite difference method near edge structure (FDMNES) program, which employs the finite difference method (FDM) to solve the Schrödinger equation and uses Green's formalism (multiple scattering) on a muffin-tin potential[37]. For Ni K-edge XANES simulations, models were constructed based on EXAFS curve fittings and optimized by DFT calculations. Final states were simulated within a Ni-centered sphere, with a radius of 7 Å, balancing accuracy and computational time.

### Computational methods

Vienna ab initio Simulation Package (VASP) package within the DFT framework was used to perform theoretical calculations[38]. Projector Augmented Wave (PAW) method was utilized to describe the interaction between ions and electrons[39], and electron exchange and correlation energy were modeled using the Perdew-Burke-Erzenhorf (PBE) functional[40]. The density of states was calculated with the HSE06 hybrid functional. London dispersion corrections (DFT-D3) were applied to account for adsorber-slab interactions[41]. Hubbard-U corrections were used for localized d-electrons of Ni and Fe, with U-J values of 3.0 eV for Fe and 5.5 eV for Ni[42,43]. A vacuum region of 15 Å was employed to separate periodic images. Brillouin-zone integrations were performed using Monkhorst-Pack grids, with a (3 × 3 × 1) k-point grid for the supercell. A plane-wave cutoff energy of 500 eV was used, with force and energy convergence thresholds set to <0.02 eV/Å and $<10^{-5}$ eV, respectively. Herein, VASPKIT code was conducted to correct Gibbs energy[44].

H desorption energy ($E_{desorp}$) was calculated as follows:

$$E_{desorp} = E_{slab} - E_{perfect} + xE_{H_2} \tag{2}$$

where $E_{slab}$ and $E_{perfect}$ are total DFT energies of defect and perfect hydroxide systems, respectively, and $E_{H_2}$ is total energy of $H_2$.

## Data availability
Source data are provided with this paper.

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## Author contributions
T.Y. conceived and supervised the project. T.Y., L.C., and Z.J. developed the idea and designed experiments. D.W. and X.L. and H.Z. performed the catalyst synthesis and characterizations, XAFS measurements, and electrochemical experiments. L.H., T.L., X.Z., Q.L. and J.Y. conducted machine learning analysis and discussed theoretical calculations. D.W., L.C., Z.J., and T.Y. co-wrote the paper. All authors discussed the results and commented on the manuscript.

## Competing interests
The authors declare no competing interests.

## Additional information
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